

2.1 Hyperbolic functions

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

$\sinh x = (e^x - e^{-x})/2$ and $\cosh x = (e^x + e^{-x})/2$. Their identity $\cosh^2 x - \sinh^2 x = 1$ contrasts with trigonometric identities; derivatives are $\cosh$ and $\sinh$ respectively. Inverse hyperbolic functions can be written using logarithms by solving a quadratic in e^x .

Worked example: follow the reasoning

$\sinh(\ln 2) = (2 - 1/2)/2 = 3/4$ and $\cosh(\ln 2) = 5/4$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Show $\cosh^2 x - \sinh^2 x = 1$ directly from exponential definitions.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Difference of squares [1]: $((e^x + e^{-x})^2 - (e^x - e^{-x})^2)/4$ [1] = $4e^x e^{-x}/4 = 1$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.